Expected Order Statistics and Advantage: Exact Computation (Known $(\bar{r}, p)$) and Unbiased Estimation from Batches

1 Overview

We study the j-th order statistic of k samples under two regimes.

Regime A (known $(\bar{r}, p)$; sampling with replacement). There are m arms. Arm $b \in \{1, \ldots, m\}$ has deterministic value $\bar{r}_b \in \mathbb{R}$ and is sampled with probability p_b, where $p \in \Delta^{m-1}$. We draw k arms i.i.d. with replacement:

$$A_1, \ldots, A_k \overset{i.i.d.}{\sim} p, \qquad X_t := \bar{r}_{A_t}, \qquad X_{(1:k)} \le \cdots \le X_{(k:k)}.$$

We compute exactly:

$$\bar{v}_j := \mathbb{E}[X_{(j:k)}], \qquad \bar{q}_{b,j} := \mathbb{E}[X_{(j:k)} \mid A_1 = b], \qquad \bar{a}_{b,j} := \bar{q}_{b,j} - \bar{v}_j.$$

Conditioning on the *first draw* yields the exact mixture identity

$$\bar{v}_j = \sum_{b=1}^{m} p_b \, \bar{q}_{b,j}, \qquad \text{equivalently} \qquad \sum_{b=1}^{m} p_b \, \bar{a}_{b,j} = 0.$$

Regime B (unknown p; i.i.d. batch; unbiased estimation). We observe a batch of size N drawn i.i.d. from the same arm distribution:

$$A^{(1)}, \ldots, A^{(N)} \overset{i.i.d.}{\sim} p, \qquad r_i := q_{A^{(i)}}.$$

We build unbiased estimators of $\bar{v}_j$ via U-statistics, and we also compute a *batch advantage* $a_{i,j}$ for each datapoint index i using include-one and leave-one-out subset expectations on the realized batch. A key equivalence proved in this note connects the exact, known-$(\bar{r}, p)$ advantage to a batch-computable advantage. In the batch regime, for each datapoint index $i \in \{1, \ldots, N\}$ we define the *batch advantage*

$$a_{i,j} := q_{i,j} - v_j^{(-i)},$$

where $q_{i,j}$ is the expected j-th order statistic of a uniformly random size-k subset *conditioned to include* index i (batch fixed), and $v_j^{(-i)}$ is the expected j-th order statistic of a uniformly random size-k subset from the batch with index i removed.

Let $I \sim \mathrm{Unif}\{1, \ldots, N\}$ be an index drawn uniformly at random, independent of the batch, and write $A^{(I)}$ for the action label at that index. Then we show

$$\boxed{\mathbb{E}\big[a_{I,j} \mid A^{(I)} = b\big] = \bar{a}_{b,j},}$$

i.e., the expected batch advantage of a randomly selected datapoint with action b equals the exact advantage in the known-$(\bar{r}, p)$ model.

No tie collapsing. We never collapse equal values. Sorting is stable; all formulas below remain exact for expectations even with ties.

Computation. Many quantities admit a *sort* $\to$ *(vectorized) matrix operations* $\to$ *invert-sort* workflow, with substantial precomputation possible when (m, k) or (N, k) are fixed.

2 Regime A: exact computation for known $(\bar{r}, p)$ (with replacement)

2.1 Sorting and CDF grid (no tie collapsing)

Let π be a permutation that sorts arms by value:

$$\bar{r}_{(1)} \le \bar{r}_{(2)} \le \cdots \le \bar{r}_{(m)}, \qquad p_{(t)} := p_{\pi(t)}.$$

Define the CDF grid

$$F_0 := 0, \qquad F_t := \mathbb{P}(X \le \bar{r}_{(t)}) = \sum_{s=1}^{t} p_{(s)}, \qquad t = 1, \ldots, m.$$

When ties are present, π can be any stable sorting permutation; then F_t is still the probability mass of values $\le \bar{r}_{(t)}$, and the results below remain exact for expectations.

2.2 Unconditional order statistics: derivation and matrix form

Fix $j \in \{1, \ldots, k\}$.

Lemma 1 (Order-statistic CDF as a binomial tail)**.** *For each $t \in \{0, \ldots, m\}$,*

$$\mathbb{P}(X_{(j:k)} \le \bar{r}_{(t)}) = \mathbb{P}\big(\mathrm{Bin}(k, F_t) \ge j\big).$$

Proof. Let $Y_t := \#\{\ell \in \{1, \ldots, k\} : X_\ell \le \bar{r}_{(t)}\}$. Since the X_t are i.i.d. and $\mathbb{P}(X_\ell \le \bar{r}_{(t)}) = F_t$, we have $Y_t \sim \mathrm{Bin}(k, F_t)$. The event $\{X_{(j:k)} \le \bar{r}_{(t)}\}$ holds iff at least j of the k values are $\le \bar{r}_{(t)}$, i.e. iff $Y_t \ge j$. This yields the claim. $\square$

Define the tail table

$$T^{(k)}_{t,j} := \mathbb{P}\big(\mathrm{Bin}(k, F_t) \ge j\big), \qquad t = 0, \ldots, m, \quad j = 1, \ldots, k. \tag{1}$$

By Lemma 1, $T^{(k)}_{t,j} = \mathbb{P}(X_{(j:k)} \le \bar{r}_{(t)})$. Because $X_{(j:k)}$ takes values among $\{\bar{r}_{(1)}, \ldots, \bar{r}_{(m)}\}$, its probability mass on the sorted grid is obtained by differencing the CDF:

$$W^{\mathrm{wr}}_{t,j} := \mathbb{P}(X_{(j:k)} = \bar{r}_{(t)}) = T^{(k)}_{t,j} - T^{(k)}_{t-1,j}, \qquad t = 1, \ldots, m. \tag{2}$$

Then

$$\bar{v}_j = \mathbb{E}[X_{(j:k)}] = \sum_{t=1}^{m} \bar{r}_{(t)}\, W^{\mathrm{wr}}_{t,j}. \tag{3}$$

Matrix form. Let $\bar{r}_{(\cdot)} := (\bar{r}_{(1)}, \dots, \bar{r}_{(m)})^\top \in \mathbb{R}^m$, let $T^{(k)} \in \mathbb{R}^{(m+1)\times k}$ collect (1), and let $D \in \mathbb{R}^{m\times(m+1)}$ be the forward-difference operator

$$D_{t,t} = 1, \quad D_{t,t-1} = -1, \quad \text{all other entries } 0.$$

Then (2) is exactly $W^{\mathrm{wr}} = DT^{(k)}$, and stacking (3) for all j yields

$$\boxed{\bar{v} := (\bar{v}_1, \dots, \bar{v}_k) = \bar{r}_{(\cdot)}^\top W^{\mathrm{wr}} = \bar{r}_{(\cdot)}^\top DT^{(k)}.} \tag{4}$$

2.3 Conditional-on-first-draw order statistics $\bar{q}_{b,j}$

Fix an arm $b \in \{1, \dots, m\}$ and a statistic index $j \in \{1, \dots, k\}$. Conditioning on $A_1 = b$ pins $X_1 = \bar{r}_b$ and leaves $X_2, \dots, X_k$ i.i.d. with the same marginal distribution as X.

For each $t \in \{0, \dots, m\}$ define

$$\delta_{b,t} := \mathbf{1}\{\bar{r}_b \le \bar{r}_{(t)}\}.$$

Lemma 2 (Conditional CDF given $A_1 = b$)**.** *For each $t \in \{0, \dots, m\}$,*

$$\mathbb{P}(X_{(j:k)} \le \bar{r}_{(t)} \mid A_1 = b) = \mathbb{P}\big(\mathrm{Bin}(k-1, F_t) \ge j - \delta_{b,t}\big),$$

with the conventions $\mathbb{P}(\mathrm{Bin}(\cdot) \ge s) = 1$ for $s \le 0$ and 0 for $s > k-1$.

Proof. Let $Y_t := \#\{\ell \in \{2, \dots, k\} : X_\ell \le \bar{r}_{(t)}\}$ count the number of the remaining $k-1$ draws that are at most $\bar{r}_{(t)}$. Conditioned on $A_1 = b$, the variables $X_2, \dots, X_k$ remain i.i.d. with $\mathbb{P}(X_\ell \le \bar{r}_{(t)}) = F_t$, hence $Y_t \sim \mathrm{Bin}(k-1, F_t)$. Including $X_1 = \bar{r}_b$ adds $\delta_{b,t}$ additional "success" if $\bar{r}_b \le \bar{r}_{(t)}$, so the total count of values $\le \bar{r}_{(t)}$ equals $\delta_{b,t} + Y_t$. As in Lemma 1, the event $\{X_{(j:k)} \le \bar{r}_{(t)}\}$ is equivalent to having at least j values $\le \bar{r}_{(t)}$, i.e. $\delta_{b,t} + Y_t \ge j$. Rearranging yields $Y_t \ge j - \delta_{b,t}$, proving the claim. □

Define the conditional tail table

$$T_{t,j}^{(b)} := \mathbb{P}(X_{(j:k)} \le \bar{r}_{(t)} \mid A_1 = b) = \mathbb{P}\big(\mathrm{Bin}(k-1, F_t) \ge j - \delta_{b,t}\big), \qquad t = 0, \dots, m, \quad j = 1, \dots, k. \tag{5}$$

As before, the conditional pmf is obtained by differencing in t:

$$W_{t,j}^{(b)} := \mathbb{P}(X_{(j:k)} = \bar{r}_{(t)} \mid A_1 = b) = T_{t,j}^{(b)} - T_{t-1,j}^{(b)}, \qquad t = 1, \dots, m, \quad j = 1, \dots, k. \tag{6}$$

and therefore

$$\bar{q}_{b,j} := \mathbb{E}[X_{(j:k)} \mid A_1 = b] = \sum_{t=1}^{m} \bar{r}_{(t)} \, W_{t,j}^{(b)}. \tag{7}$$

Matrix form. Let $W^{(b)} \in \mathbb{R}^{m\times k}$ collect $W_{t,j}^{(b)}$ across j, then

$$\boxed{\bar{q}_b := (\bar{q}_{b,1}, \dots, \bar{q}_{b,k}) = \bar{r}_{(\cdot)}^\top W^{(b)} = \bar{r}_{(\cdot)}^\top DT^{(b)}.} \tag{8}$$

Shared precomputation. Let

$$T_{t,j}^{(k-1)} := \mathbb{P}(\mathrm{Bin}(k-1, F_t) \geq j), \qquad t = 0, \dots, m, \quad j = 0, 1, \dots, k,$$

where $T_{t,0}^{(k-1)} = 1$. Then Lemma 2 implies a two-case splice:

$$T_{t,j}^{(b)} = \begin{cases} T_{t,j}^{(k-1)}, & \delta_{b,t} = 0, \\ T_{t,j-1}^{(k-1)}, & \delta_{b,t} = 1, \end{cases}$$

Thus, after sorting and forming the grid F_t, one can compute $T^{(k)}$ and $T^{(k-1)}$ once and then obtain each $T^{(b)}$ (hence $\bar{q}_b$) by a simple step-function splice, followed by differencing and a dot with $\bar{r}_{(\cdot)}$.

2.4 Mixture identity and advantage

Proposition 1 (Mixture identity for conditioning on the first draw). *For each $j \in \{1, \dots, k\}$,*

$$\bar{v}_j = \sum_{b=1}^{m} p_b \, \bar{q}_{b,j}.$$

Proof. By the law of total expectation and the fact that $\{A_1 = b\}_{b=1}^m$ partitions the sample space,

$$\mathbb{E}[X_{(j:k)}] = \sum_{b=1}^{m} \bar{q}_{b,j} \; \mathbb{P}(A_1 = b) = \sum_{b=1}^{m} \bar{q}_{b,j} \, p_b.$$

□

Define the advantage of arm b:

$$\boxed{\bar{a}_{b,j} := \bar{q}_{b,j} - \bar{v}_j, \qquad a_b := (\bar{q}_{b,1} - \bar{v}_1, \dots, \bar{q}_{b,k} - \bar{v}_k) \in \mathbb{R}^k.} \tag{9}$$

Then $\sum_b p_b \bar{a}_{b,j} = 0$ for every j.

L-statistics. For weights $\alpha \in \mathbb{R}^k$, define the L-statistic $T = \sum_{j=1}^k \alpha_j X_{(j:k)} = \alpha^\top (X_{(1:k)}, \dots, X_{(k:k)})$. Then $\mathbb{E}[T] = \alpha^\top \bar{v}$, $\bar{q}_b^{(\alpha)} := \alpha^\top \bar{q}_b$, and $a_b^{(\alpha)} := \bar{q}_b^{(\alpha)} - \alpha^\top \bar{v}$.

Lemma 3 (Random subsample of i.i.d. observations). *Let $Z_1, \dots, Z_M$ be i.i.d. random variables with common law P. Let $J = \{J_1, \dots, J_s\}$ be a uniformly random subset of $\{1, \dots, M\}$ of size s, chosen independently of $Z_{1:M}$. Then the unordered multiset $\{Z_{J_1}, \dots, Z_{J_s}\}$ has the same distribution as $\{Z_1, \dots, Z_s\}$. Equivalently, for any symmetric function $h : \mathbb{R}^s \to \mathbb{R}$,*

$$\mathbb{E}\big[h(Z_{J_1}, \dots, Z_{J_s})\big] = \mathbb{E}\big[h(Z_1, \dots, Z_s)\big].$$

Proof. Condition on J. Because $(Z_1, \dots, Z_M)$ are i.i.d., the joint distribution of $(Z_{J_1}, \dots, Z_{J_s})$ (for any fixed ordered listing of J) equals that of $(Z_1, \dots, Z_s)$. Since J is chosen independently of $Z_{1:M}$, averaging over J preserves this distributional equality for any symmetric function h. □

3 Regime B: unbiased estimation from an i.i.d. batch (unknown p)

Throughout this section, the probability distribution p is unknown; we observe an i.i.d. batch and estimate the targets from Regime A.

3.1 Unbiased estimation of $\bar{v}_j$ via U-statistics

Observe N i.i.d. samples $A^{(1)}, \ldots, A^{(N)} \sim p$ and values $r_i = \bar{r}_{A^{(i)}}$. Fix $k \le N$ and define the U-statistic estimator

$$v_j := \frac{1}{\binom{N}{k}} \sum_{S \subseteq \{1,\ldots,N\}:\, |S|=k} \big(r_S\big)_{(j:k)}, \tag{10}$$

where $(r_S)_{(j:k)}$ is the j-th order statistic of the multiset $\{r_i : i \in S\}$.

Theorem 1 (Unbiasedness of the U-statistic estimator)**.** *If $r_1, \ldots, r_N$ are i.i.d. with the same distribution as $X = \bar{r}_A$ under $A \sim p$, then for each j,*

$$\mathbb{E}[v_j] = \bar{v}_j.$$

Proof. Let S be a uniformly random size-k subset of $\{1, \ldots, N\}$, independent of the values. Then (10) is exactly $v_j = \mathbb{E}[(r_S)_{(j:k)} \mid r_{1:N}]$. Taking expectation over the batch,

$$\mathbb{E}[v_j] = \mathbb{E}[(r_S)_{(j:k)}].$$

By Lemma 3, because $r_1, \ldots, r_N$ are i.i.d. and S is uniform and independent, the unordered multiset $\{r_i : i \in S\}$ has the same distribution as $\{X_1, \ldots, X_k\}$ where $X_1, \ldots, X_k$ are i.i.d. copies of X. Therefore $(r_S)_{(j:k)}$ has the same distribution as $X_{(j:k)}$, and $\mathbb{E}[(r_S)_{(j:k)}] = \mathbb{E}[X_{(j:k)}] = \bar{v}_j$. □

3.2 Sort + weights derivation (computing all j)

Let $r_{(1)} \le \cdots \le r_{(N)}$ be the sorted values and $r := (r_{(1)}, \ldots, r_{(N)})^\top$. For each $m \in \{1, \ldots, N\}$ and $j \in \{1, \ldots, k\}$,

$$\boxed{W_{m,j} = \mathbb{P}\big((r_S)_{(j:k)} = r_{(m)} \mid r_{1:N}\big) = \frac{\binom{m-1}{j-1}\binom{N-m}{k-j}}{\binom{N}{k}}.} \tag{11}$$

Consequently,

$$\boxed{v := (v_1, \ldots, v_k) = r^\top W.} \tag{12}$$

Derivation of (11)*.* Fix m and j. The event $\{(r_S)_{(j:k)} = r_{(m)}\}$ occurs iff: (i) $r_{(m)}$ is included in S, (ii) exactly $j-1$ of the selected indices are from the $m-1$ smaller values $\{r_{(1)}, \ldots, r_{(m-1)}\}$, and (iii) the remaining $k-j$ indices are from the $N-m$ larger values $\{r_{(m+1)}, \ldots, r_{(N)}\}$. The number of such subsets is $\binom{m-1}{j-1}\binom{N-m}{k-j}$ out of the $\binom{N}{k}$ total size-k subsets. □

L-statistics. If α is fixed, precompute $w^{(\alpha)} := W\alpha \in \mathbb{R}^N$ (depends only on N, k, α), and compute $\widehat{T} = r^\top w^{(\alpha)}$ after sorting.

3.3 Include-one and leave-one-out expectations on the realized batch

These are defined *given a realized dataset*; expectation is with respect to random subset selection without replacement.

Let S be a uniform size-k subset of $\{1, \ldots, N\}$. For each index i and each j, define

$$q_{i,j} := \mathbb{E}\big[(r_S)_{(j:k)} \mid i \in S,\ r_{1:N}\big], \tag{13}$$

$$v_j^{(-i)} := \mathbb{E}\big[(r_{S'})_{(j:k)} \mid r_{1:N} \text{ with index } i \text{ removed}\big], \tag{14}$$

where S' is a uniform size-k subset of the remaining $N-1$ indices. Define the *batch advantage*

$$\boxed{a_{i,j} := q_{i,j} - v_j^{(-i)}.} \tag{15}$$

All formulas are simplest in sorted-rank space; invert the sorting permutation to attach outputs to the original indices.

3.3.1 Explicit include-one weights A, B, C (proved)

Work in *sorted-rank* space. Let $r_{(1)} \le \cdots \le r_{(N)}$ be the sorted observed values and let $t \in \{1, \ldots, N\}$ denote a particular rank in this sorted list.

Conditioning on including rank t means we force $r_{(t)}$ into the size-k subset and choose the remaining $k-1$ indices uniformly from the other $N-1$ ranks.

Lemma 4 (Include-one pmf weights)**.** *Fix $j \in \{1, \ldots, k\}$ and an included rank $t \in \{1, \ldots, N\}$. For each $m \in \{1, \ldots, N\}$,*

$$\mathbb{P}\big((r_S)_{(j:k)} = r_{(m)} \mid t \in S,\ r_{1:N}\big) = \begin{cases} \dfrac{\binom{m-1}{j-1}\binom{N-m-1}{k-j-1}}{\binom{N-1}{k-1}}, & m < t, \\ \dfrac{\binom{t-1}{j-1}\binom{N-t}{k-j}}{\binom{N-1}{k-1}}, & m = t, \\ \dfrac{\binom{m-2}{j-2}\binom{N-m}{k-j}}{\binom{N-1}{k-1}}, & m > t. \end{cases}$$

Proof. We count size-k subsets that contain rank t and have $r_{(m)}$ as the j-th order statistic.

Case $m < t$. The forced element $r_{(t)}$ is larger than $r_{(m)}$, so it must be among the values above $r_{(m)}$. We must include $r_{(m)}$; choose $j-1$ from the $m-1$ smaller values; and choose the remaining $k-j-1$ (excluding the forced rank t and excluding m) from the values above $r_{(m)}$. After removing t and m, there are $N-m-1$ values above $r_{(m)}$, giving $\binom{m-1}{j-1}\binom{N-m-1}{k-j-1}$ subsets.

Case $m = t$. The forced element is itself the j-th order statistic, so we choose $j-1$ from the $t-1$ values below it and $k-j$ from the $N-t$ values above it. Count: $\binom{t-1}{j-1}\binom{N-t}{k-j}$.

Case $m > t$. The forced element $r_{(t)}$ is below $r_{(m)}$, so it must be counted among the $j-1$ values below $r_{(m)}$. We include $r_{(m)}$; choose the remaining $j-2$ values below $r_{(m)}$ from the $m-2$ values in $\{1, \ldots, m-1\} \setminus \{t\}$; and choose $k-j$ from the $N-m$ values above $r_{(m)}$. Count: $\binom{m-2}{j-2}\binom{N-m}{k-j}$.

The conditional sample space (size-k subsets containing rank t) has cardinality $\binom{N-1}{k-1}$, so dividing by $\binom{N-1}{k-1}$ yields the probabilities. $\square$

Define three precomputable matrices $A, B, C \in \mathbb{R}^{N \times k}$ by

$$A_{m,j} := \frac{\binom{m-1}{j-1}\binom{N-m-1}{k-j-1}}{\binom{N-1}{k-1}}, \tag{16}$$

$$B_{m,j} := \frac{\binom{m-1}{j-1}\binom{N-m}{k-j}}{\binom{N-1}{k-1}}, \tag{17}$$

$$C_{m,j} := \frac{\binom{m-2}{j-2}\binom{N-m}{k-j}}{\binom{N-1}{k-1}}. \tag{18}$$

Then Lemma 4 says: for included rank t, the pmf over m is $A_{m,j}$ for $m < t$, $B_{t,j}$ for $m = t$, and $C_{m,j}$ for $m > t$.

Include-one expectation (rank space). Define

$$q_{t,j} := \mathbb{E}\big[(r_S)_{(j:k)} \mid t \in S,\ r_{1:N}\big].$$

Then

$$q_{t,j} = \sum_{m<t} r_{(m)} A_{m,j} + r_{(t)} B_{t,j} + \sum_{m>t} r_{(m)} C_{m,j}. \tag{19}$$

Mapping back to the original datapoint index i via the sorting permutation yields $q_{i,j}$.

3.3.2 Explicit leave-one-out weights W^- (proved)

Remove rank t and sample a uniform size-k subset from the remaining $N-1$ ranks. Let $W^- \in \mathbb{R}^{(N-1)\times k}$ be the unconditional weights for a population of size $N-1$:

$$W^-_{\ell,j} = \frac{\binom{\ell-1}{j-1}\binom{(N-1)-\ell}{k-j}}{\binom{N-1}{k}}, \qquad \ell = 1,\dots,N-1. \tag{20}$$

After removing rank t, the reduced sorted list $r^{(-t)}_{(1)} \le \cdots \le r^{(-t)}_{(N-1)}$ satisfies

$$r^{(-t)}_{(\ell)} = \begin{cases} r_{(\ell)}, & \ell < t, \\ r_{(\ell+1)}, & \ell \ge t. \end{cases}$$

Therefore, defining

$$v^{(-t)}_j := \mathbb{E}\big[(r_{S'})_{(j:k)} \mid \text{rank } t \text{ removed},\ r_{1:N}\big],$$

we obtain

$$v^{(-t)}_j = \sum_{m<t} r_{(m)}\, W^-_{m,j} + \sum_{m>t} r_{(m)}\, W^-_{m-1,j}. \tag{21}$$

Mapping back to the original datapoint index i yields $v^{(-i)}_j$.

3.3.3 Scalable $O(Nk)$ computation via prefix/suffix sums (derived)

Equations (19) and (21) can be evaluated for all t and j in $O(Nk)$ time using cumulative sums.

Include-one. Let $R := r\,\mathbf{1}_k^\top \in \mathbb{R}^{N\times k}$ replicate the sorted values across k columns and define

$$P_A := R \odot A, \qquad P_C := R \odot C,$$

with Hadamard product $\odot$. Let cumsum denote cumulative sum down rows. Then, with the convention that $\mathrm{cumsum}(P)_{0,:} = 0$,

$$\sum_{m<t} r_{(m)} A_{m,:} = \mathrm{cumsum}(P_A)_{t-1,:}, \qquad \sum_{m>t} r_{(m)} C_{m,:} = \mathrm{cumsum}(P_C)_{N,:} - \mathrm{cumsum}(P_C)_{t,:}.$$

Thus $q_{t,:} = (q_{t,1},\dots,q_{t,k})$ for all t can be obtained by two cumulative sums and a few rowwise operations.

Leave-one-out. Let $u := r_{(1:N-1)}$ and $v := r_{(2:N)}$ (views of the sorted vector). Form $P_1 := u\,\mathbf{1}_k^\top \odot W^-$ and $P_2 := v\,\mathbf{1}_k^\top \odot W^-$. Then for each t,

$$\sum_{m<t} r_{(m)} W^-_{m,:} = \mathrm{cumsum}(P_1)_{t-1,:}, \qquad \sum_{m>t} r_{(m)} W^-_{m-1,:} = \mathrm{cumsum}(P_2)_{N-1,:} - \mathrm{cumsum}(P_2)_{t-1,:}.$$

This yields $v^{(-t)}$ for all t in $O(Nk)$ time.

Batch advantage. In sorted-rank space, define

$$A_{t,j} := q_{t,j} - v_j^{(-t)},$$

and map back to the original datapoint index i via the inverse sorting permutation to obtain $a_{i,j}$.

3.3.4 Tiny-N linear-operator forms (optional)

When N is tiny (e.g. $N \le 10$), it can be convenient to view (19)–(21) as explicit linear operators applied to the sorted vector r.

Let $U, L \in \{0,1\}^{N\times N}$ be the strict upper and lower triangular masks:

$$U_{m,t} := \mathbf{1}\{m < t\}, \qquad L_{m,t} := \mathbf{1}\{m > t\}.$$

Include-one operator for fixed j. Fix j and let $a, b, c \in \mathbb{R}^N$ be the j-th columns of A, B, C. Then the include-one operator $W^{\mathrm{inc}}_{(j)} \in \mathbb{R}^{N\times N}$ defined by

$$W^{\mathrm{inc}}_{(j)} := (a\mathbf{1}_N^\top) \odot U + \mathrm{diag}(b) + (c\mathbf{1}_N^\top) \odot L \tag{22}$$

satisfies $(r^\top W^{\mathrm{inc}}_{(j)})_t = q_{t,j}$ for every t.

Leave-one-out operator for fixed j. Fix j and let $u \in \mathbb{R}^{N-1}$ be the j-th column of W^-; define $u^\uparrow := (u_1, \ldots, u_{N-1}, 0)^\top$ and $u^\downarrow := (0, u_1, \ldots, u_{N-1})^\top$. Then

$$W^{\mathrm{loo}}_{(j)} := (u^\uparrow\mathbf{1}_N^\top) \odot U + (u^\downarrow\mathbf{1}_N^\top) \odot L \tag{23}$$

satisfies $(r^\top W^{\mathrm{loo}}_{(j)})_t = v_j^{(-t)}$ for every t. Hence the advantage operator for fixed j is $W^{\mathrm{adv}}_{(j)} := W^{\mathrm{inc}}_{(j)} - W^{\mathrm{loo}}_{(j)}$.

L-statistic collapse. For fixed $\alpha \in \mathbb{R}^k$, define $\tilde a := A\alpha$, $\tilde b := B\alpha$, $\tilde c := C\alpha$, and $u := W^-\alpha$. Then one obtains single $N\times N$ operators $W^{\mathrm{inc}}_{(\alpha)}, W^{\mathrm{loo}}_{(\alpha)}$ by replacing a, b, c and u above, and $r^\top(W^{\mathrm{inc}}_{(\alpha)} - W^{\mathrm{loo}}_{(\alpha)})$ yields the advantage for the L-statistic.

3.4 Equivalence of advantages (batch → known $(\bar r, p)$) with full proof

Let I be uniformly random in $\{1, \ldots, N\}$, independent of the dataset. Fix an arm b. Recall the known-$(\bar r, p)$ advantage $\bar a_{b,j} = \bar q_{b,j} - \bar v_j$ from (9).

Theorem 2 (Advantage equivalence). *For each $j \in \{1, \ldots, k\}$,*

$$\boxed{\mathbb{E}\big[a_{I,j} \mid A^{(I)} = b\big] = \bar a_{b,j}.} \tag{24}$$

Proof. We prove that

$$\mathbb{E}\big[q_{I,j} \mid A^{(I)} = b\big] = \bar q_{b,j}, \qquad \mathbb{E}\big[v_j^{(-I)} \mid A^{(I)} = b\big] = \bar v_j,$$

and subtract.

Step 1: include-one term. Condition on the event $A^{(I)} = b$. Then $r_I = \bar{r}_b$ is fixed. Given the realized dataset, $q_{I,j}$ is the expectation of the j-th order statistic of a size-k subset conditioned to include index I, i.e. it is the expectation of

$$g_j\big(r_I, r_{J_1}, \dots, r_{J_{k-1}}\big),$$

where $\{J_1, \dots, J_{k-1}\}$ is a uniformly random $(k-1)$-subset of the remaining indices $\{1, \dots, N\} \setminus \{I\}$.

Let $Z_1, \dots, Z_{N-1}$ denote the multiset of values in $\{r_\ell : \ell \neq I\}$. Conditional on $A^{(I)} = b$, these Z_ℓ are still i.i.d. draws from the marginal distribution of $X = \bar{r}_A$ under $A \sim p$ (because the dataset is i.i.d. and I is independent of the data). Moreover, by Lemma 3, a uniform subset of size $k-1$ drawn without replacement from $Z_{1:N-1}$ has the same unconditional distribution as $(k-1)$ i.i.d. draws from the same law. Thus the random multiset $\{Z_{J_1}, \dots, Z_{J_{k-1}}\}$ is distributed as $\{X_2, \dots, X_k\}$, where $X_2, \dots, X_k$ are i.i.d. copies of X independent of $\bar{r}_b$. Therefore,

$$\mathbb{E}\big[q_{I,j} \mid A^{(I)} = b\big] = \mathbb{E}\big[g_j(\bar{r}_b, X_2, \dots, X_k)\big] = \bar{q}_{b,j}.$$

Step 2: leave-one-out term. Condition on $A^{(I)} = b$. Removing index I leaves $N-1$ values which remain i.i.d. with the law of X. The leave-one-out quantity $v_j^{(-I)}$ is the U-statistic expectation (over uniform size-k subsets) computed on these $N-1$ points. By Theorem 1 applied with $N-1$ samples, its expectation equals the target $\bar{v}_j$:

$$\mathbb{E}\big[v_j^{(-I)} \mid A^{(I)} = b\big] = \bar{v}_j.$$

Step 3: subtract. By definition $a_{I,j} = q_{I,j} - v_j^{(-I)}$, hence

$$\mathbb{E}\big[a_{I,j} \mid A^{(I)} = b\big] = \bar{q}_{b,j} - \bar{v}_j = \bar{a}_{b,j}.$$

$\square$

L-statistics. For any fixed $\alpha \in \mathbb{R}^k$, define $A_I^{(\alpha)} := \sum_{j=1}^k \alpha_j a_{I,j}$ and $a_b^{(\alpha)} := \alpha^\top a_b$. Linearity of expectation and Theorem 2 give

$$\mathbb{E}\big[A_I^{(\alpha)} \mid A^{(I)} = b\big] = a_b^{(\alpha)}.$$

4 Precomputation and workflow

Known $(\bar{r}, p)$ (exact). For fixed (m, k):

- Precompute $\binom{k}{s}$ and $\binom{k-1}{s}$ for $s = 0, \dots, k$ (or use stable log-factorial tables).
- For each new $(\bar{r}, p)$: stable sort by $\bar{r}$ (permute p), compute the grid F_t, compute tail tables $T^{(k)}$ and $T^{(k-1)}$ via binomial pmfs and reverse cumsums or special functions, difference in t, then multiply by $\bar{r}_{(\cdot)}$.
- Conditional $\bar{q}_b$ uses only splicing of $T^{(k-1)}$ based on $\delta_{b,t} = \mathbf{1}\{\bar{r}_b \le \bar{r}_{(t)}\}$, followed by the same difference and dot.

Unknown p (batch). For fixed (N, k):

- Precompute $W(N, k)$ in (11). For include/loo/advantage, precompute A, B, C in (16) and W^- in (20).
- For each batch: sort r, apply the precomputed operators (matrix multiply and/or cumsums), and invert the sort if per-index outputs are required.
- If a fixed L-statistic α is used repeatedly, collapse the precomputed matrices once (e.g. $W\alpha$, $A\alpha$, $B\alpha$, $C\alpha$, $W^-\alpha$) to reduce per-batch computation.